

FIRST. / LAST.

Make row order part of the request

01 / THE SAS EXAMPLE

```
data test;
  input id group $ value;
  datalines;
1 A 10
2 A 20
3 B 30
4 B 40
5 B 50
6 C 60
;
run;
data flagged;
  set test;
  by group;
  first_flag = first.group;
  last_flag = last.group;
run;
proc print data=flagged noobs;
run;
```

REPORTED TEST SETUP / 21 SEP 2026
Claude Sonnet 4.5 / AWS Bedrock GovCloud
Claude Code 2.1.278 / effortLevel=high / prompt v0.1.0

ATTACH SOURCE OR PASTE IT AFTER THE PROMPT. START FRESH EACH TIME.

WHAT YOU WILL PRACTICE

State the ordering assumptions that define a row-based result.

These six records are sorted by group. FIRST.group and LAST.group mark each group's beginning and end. Group C has one member, so its row receives both flags.

EXPECTED RESULT

id	group	value	first_flag	last_flag
1	A	10	1	0
2	A	20	0	1
3	B	30	1	0
4	B	40	0	0
5	B	50	0	1
6	C	60	1	1

Scope: one nonmissing grouping variable and the supplied sorted rows. Both flags are 1 for a one-row group. All six rows remain in the output.

Reported outputs: A 3/3 | B 3/3. Compliance: see p. 3.

Reference checked in Python; SAS execution pending.

Prompt A: leave choices open

High degrees of freedom / Notice the decisions the model makes for you.

COPY THIS PROMPT + THE SAS EXAMPLE

Convert this SAS code to Python.
Show code in this response. Do not create or run files, or read existing Python files.

REPORTED TRIAL RESULTS / 21 SEP 2026

All A runs chose group positions

Run	Method	Output match
A1	groupby().cumcount()	Yes
A2	groupby().cumcount()	Yes
A3	groupby().cumcount()	Yes

All three report correct flags; none included assertions.
Position 0 marks first; position size - 1 marks last.
Agreement in three reports does not guarantee future runs.

SOURCES: results-20260921.csv; trial-log-20260921.md; evidence-review-20260922.md

WHY CHOICES REMAIN OPEN

- 1

The source supplies the order

The BY statement and sorted rows support a correct translation even with a short request.
- 2

More than one method works here

Group positions and adjacent-row comparisons produce the same flags for these sorted, nonmissing groups.
- 3

Correct flags still need checking

Keep all six rows in order. Test both lists, including interior rows and the one-row group C.

ANTI-PATTERN: "KEEP FIRST AND LAST"

That asks for filtering. This program adds flags to every row. Filtering group boundaries would discard row 4 and change the task.

Verify six ordered rows, not just group boundaries.

Prompt B: make the choices explicit

Low degrees of freedom / State the behavior, then ask for a check.

COPY THIS PROMPT + THE SAME SAS EXAMPLE

```
Convert this SAS code to Python using pandas.
Show code in this response. Do not create or run files, or read
existing Python files.

- Recreate id, group, and value. Keep all six rows in their order.
- These rows are already sorted by group, with no missing groups.
- Add first_flag and last_flag as integer 0/1 columns.
- first_flag is 1 on the first row or when group differs from
  the previous row. Compare with group.shift(1).
- last_flag is 1 on the last row or when group differs from
  the next row. Compare with group.shift(-1).
- Handle the first and last rows explicitly. Do not filter rows
  or reorder records within groups.
- Include assertions for id = [1, 2, 3, 4, 5, 6],
  first_flag = [1, 0, 1, 0, 0, 1], and
  last_flag = [0, 1, 0, 0, 1, 1].
- Return one complete script. Compute the flags from group;
  do not hard-code them. Explain why row 6 has both flags.
```

WHY THESE DETAILS HELP

- 1

Preserve rows and their sequence
One sorted, nonmissing group key; keep every row. Group C must receive both flags.
- 2

Make endpoint handling explicit
B1 includes endpoint checks. B2/B3 rely on missing comparisons, omitting this prompt requirement.
- 3

Separate output from compliance
All B outputs and assertions are reported as passing. Only B1 shows explicit endpoint handling.

REPORTED B RUNS / REVIEWED EXCERPTS

Run	Output / assertions	Explicit endpoints
B1	Yes / passed	Yes
B2	Yes / passed	No
B3	Yes / passed	No

CHECK IT YOURSELF
Check both lists, row order and explicit endpoints.
Nullable strings can propagate NA at shifted endpoints.